



The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	STA2002H
Course title (English)	Honours Probability and Statistics II
Course title (Chinese)	概率及统计荣誉课程 (二)
Units	3
Description (English)	This course is designed to introduce the basic concepts, methods, and related theoretical results regarding the statistics discipline. It also helps students learn how to design, implement, and analyze an estimator. Topics include: 1) Point estimation: bias-variance trade-off; maximum likelihood estimation; method of moments; plug-in estimation. 2) Interval estimation: concept of confidence interval; pivotal statistics; Confidence interval for various estimators. 3) Hypothesis testing: concept of testing statistical hypothesis; P-value; Type 1 & 2 error; power of a test; one-sample & two-sample test; testing theory; Parametric & Non-parametric test. 4) Simple linear regression: concept of regression; linear regression; least-square fitting. 5) Bayesian statistics: Prior / Posterior distribution; Bayesian inference. Students are also required to analyze data via python or R.
Description (Chinese)	本课程旨在介绍数理统计学中的基本概念, 方法及相关理论。同时, 本课程将教授学生如何构造及分析一个统计量。本课程包含如下内容: 1.点估计理论: 偏差及方差, 极大似然估计, 矩法及其他方法。2) 区间估计: 置信区间的定义, 枢轴统计量, 构造统计量的置信区间。3) 假设检验: 假设检验的定义, P值, 第一类和第二类错误, 单/双样本检验, 参数及无参数检验, 假设检验理论。4) 线性回归: 线性回归的定义, 最小二乘拟合。5) 贝叶斯统计: 先验/后验分布, 贝叶斯推断及一些实际应用。我们还将要求使用Python或R进行数据分析。

B. Corresponding course at CUHK

Please give details of the *closest* corresponding course at CUHK (as approved by CUHK



Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	STAT2002
Course title (English)	Probability and Statistics II
Course title (Chinese)	
Units	3
Description (English)	This course is designed to study some advanced concepts of probability and statistics. Topics include: Descriptive measures and regression. Estimation, maximum likelihood and Bayesian. Interval estimation. Estimation in linear regression models. Inference: hypothesis testing, significance and power, P-value, likelihood ratio tests. Analysis of Variance. Distribution free tests and goodness of fit testing.
Description (Chinese)	

2. Course requisites

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.

A. Prerequisites

STA2001H Honours Probability and Statis, or STA2001 Probability and Statistics I, or having permission from academic advisor

B. Co-requisites

N / A

C. Others

3. Learning outcomes

Upon completing this course, students will be able to:

- Understand the basic concept in statistics discipline, and know how to design / implement / analyze an estimator for a given data set.
- Use data to verify a scientific hypothesis.
- Be familiar with statistics notions, and understand procedures for statistical inference.
- Know how to perform data analysis via R and Python.



4. Course syllabus

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

The goal of this course is to introduce the concepts and methods regarding the statistics discipline; and help students know how to design / analyze an estimator. We will provide written homework & programming assignments to help achieve this goal. Topics covered will include the following:

1. Review & Introduction

- 1.1 The development of statistics
- 1.2 A review of probability
- 1.3 Probability results for a random vector

2. Point Estimation

- 2.1 Bias - Variance trade-off
- 2.2 Method of Moments and plug-in estimation
- 2.3 Maximum likelihood estimation

3. Interval Estimation

- 3.1 Confidence interval
- 3.2 Pivotal estimator
- 3.3 Build confidence interval for an estimator

4. Hypothesis testing

- 4.1 Concept of hypothesis testing
- 4.2 P - Value
- 4.3 Type 1 & 2 error
- 4.4 Power of a test
- 4.5 One-sample and Two-sample test
- 4.6 Parametric test
- 4.7 Non-parametric test

5. (Linear) regression

- 5.1 Concept of regression & linear regression
- 5.2 Least square fitting

6. Bayesian statistics

- 6.1 Prior & Posterior distribution
- 6.2 Bayesian inference

5. Assessment scheme



Component/ method	% weight
Homework Assignments	25
Midterm Exam	35
Final Exam	40

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type: **General**

Grade	Description
A/A-	EXCELLENT - exceptionally good performance and exceeding expectations in all or most of the course learning outcomes; demonstration of a solid understanding of the subject matter, the ability to analyze problems and apply extensive knowledge, and skillful use of concepts and materials to derive proper solutions.
B+/B/B-	GOOD – good performance in all course learning outcomes and exceeding expectations in some of them; demonstration of a good understanding of the subject matter and the ability to use proper concepts and materials to solve most of the problems encountered.
C+/C/C-	FAIR – adequate performance and meeting essential expectations in course learning outcomes; demonstration of adequate understanding of the subject matter and the ability to solve simple problems.
D+/D	MARGINAL – performance barely meets expectations in the course learning outcomes; demonstration of partial understanding of the subject matter and the ability to solve simple problems.
F	FAILURE – performance does not meet expectations in the essential course learning outcomes; demonstration of serious deficiencies and the need to retake the course.

7. Feedback for evaluation

Please specify forms of evaluation to generate feedback on classes from students.

- Informal feedback to the instructor and / or teaching assistant(s) through face-to-face talking, emails, office hours, etc.;
- Advisor-advisee program and program review.

8. Readings



A.Required:

1. Foundations of Statistics for Data Scientists: With R and Python, 978-0-367-74845-6, Alan Agresti and Maria Kateri., Chapman and Hall/CRC, 1, United States

9. Course components

Activity	Hours/week
Assignments	4
Lecture	3
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Overview of statistics and review of probability, multivariate random variable
8	Review & Midterm
9	Type - 1 error, Type - 2 error, Power of a test, two sample test in mean and ratio, 'Never confirm the null'
11	Nonparametric test, concept of regression linear regression, least-square fitting
12	Bayesian formula, Prior distribution, Posterior distribution, Bayesian inference
13	Furthre Bayesian statistics
14	Review of the class & Final
2 - 3	Bias-Variance trade-off, likelihood estimation, Method of moments, Plug-in Estimation
4 - 5	Concept of confidence interval, pivotal statistics, confidence interval for sample mean, confidence interval for proportion, confidence interval for likelihood estimator
6 - 7	Concept of hypothesis testing, P-value, P-value in R, Testing sample mean, testing sample proportion, Duality between test and Confidence interval, non-parametric test
10 - 11	Likelihood ratio test, Wald test, score test, non-parametric test, elementary testing theory

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

About 150 people a class and there are 2 sessions in total



12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

14. Any other information

N / A

15. Version date

Version number	2
As of (date)	2025/11/11